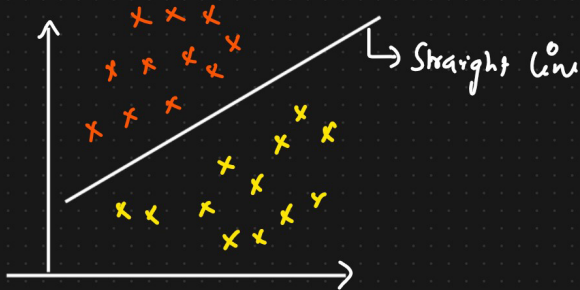
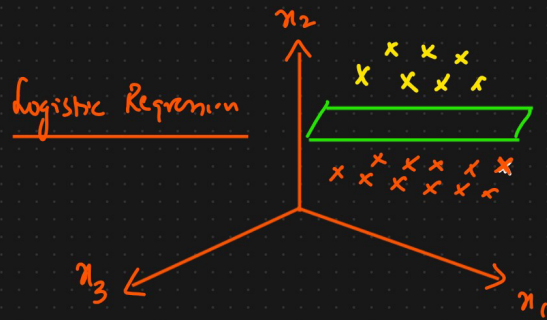


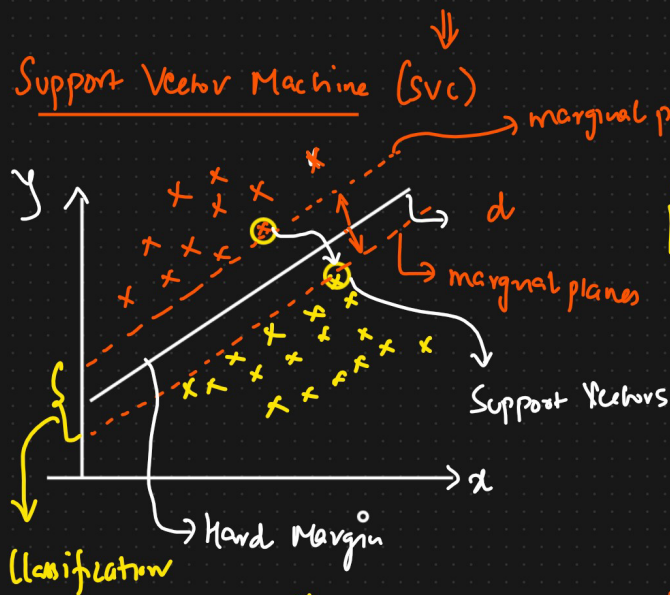
Support Vector Machines ML Algorithms.

① SVC (Support Vector Classifier)

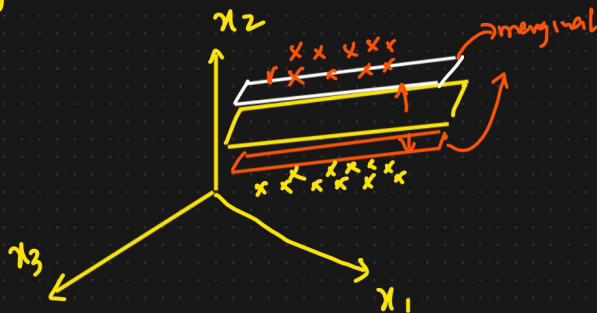
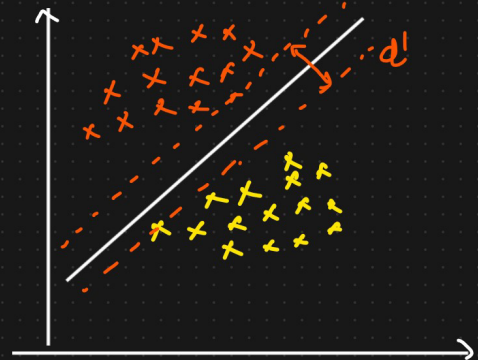
② SVR (Support Vector Regressor)



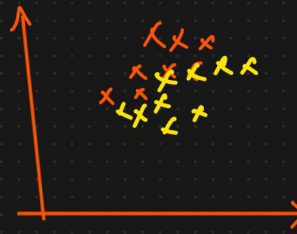
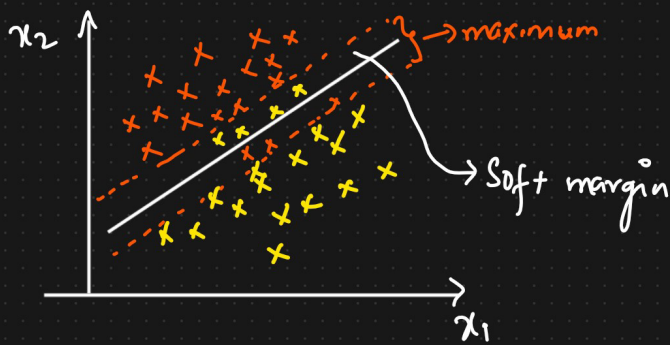
① Support Vector Machine (SVC)



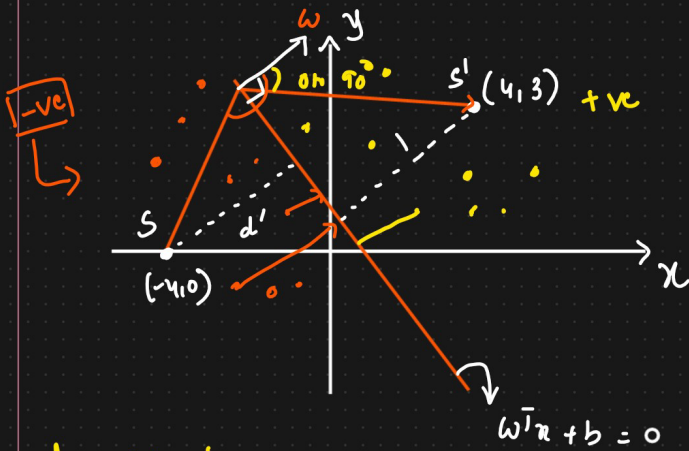
$$|d > d'|$$



Soft Margin And Hard Margin In SVM



Support Vector Machines (SVC) Maths Intuition



$$ax + by + c = 0$$

↓

$$w_1 x_1 + w_2 x_2 + b = 0$$

$$\boxed{w^T x + b = 0}$$

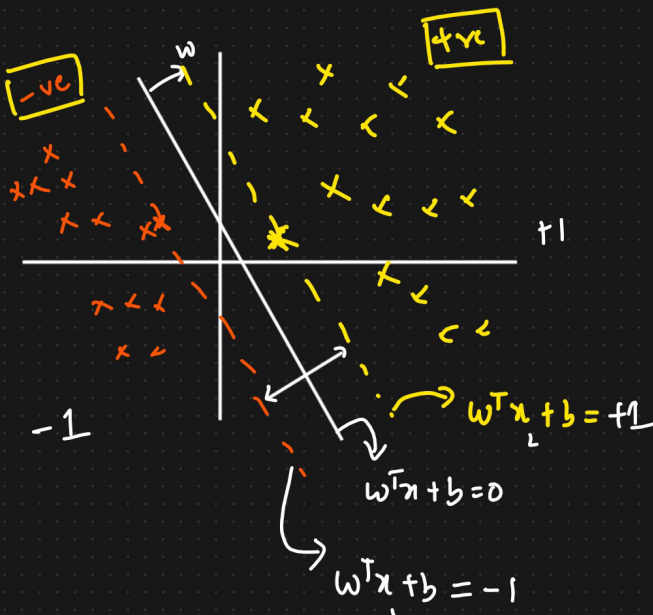
↓

$$b = 0$$

$$\boxed{w^T x = 0}$$

$d = -ve$ below plane

$d = +ve$ above plane



$$w^T x_1 + b = 1$$

$$w^T x_2 + b = -1$$

$$\begin{matrix} (-) & (-) & (+) \end{matrix}$$

$$\frac{w^T (x_1 - x_2)}{\|w\|} = \frac{+2}{\|w\|}$$

Unit vector { Magnitude of the vector is 1 }

Cost function

Maximize $\frac{2}{\|w\|}$ $\Rightarrow$ Distance between marginal plane
 w, b

Constraint such that

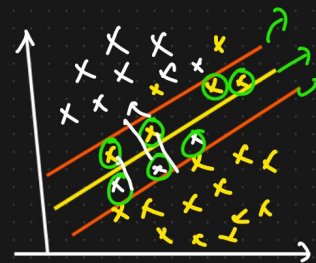
$$y_i \begin{cases} +1 & w^T x + b \geq 1 \\ -1 & w^T x + b \leq -1 \end{cases}$$

For all correct points

Constraint $\rightarrow \boxed{y_i * (w^T x + b) \geq 1}$

Maximize $\frac{2}{\|w\|}$ $\Rightarrow$ $\boxed{\text{Min } \frac{\|w\|}{2}}$
 w, b

$C_i = 6$ ✓



Cost function of SVM (svc)

Min $\frac{\|w\|}{2} + \sum_{i=1}^n C_i \xi_i \Rightarrow$ Hinge Loss

$\Downarrow$

Soft margin

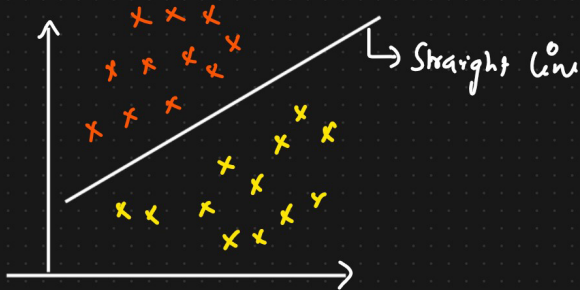
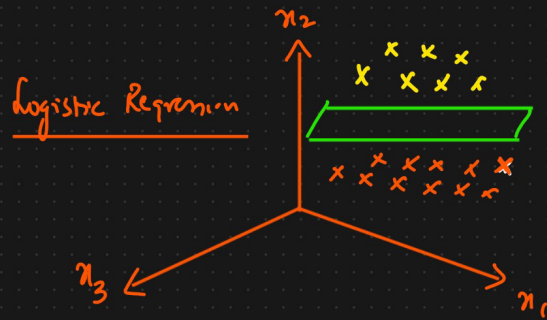
{ how many points we want to avoid misclassification }

{ distance of the incorrect data points from the marginal plane }

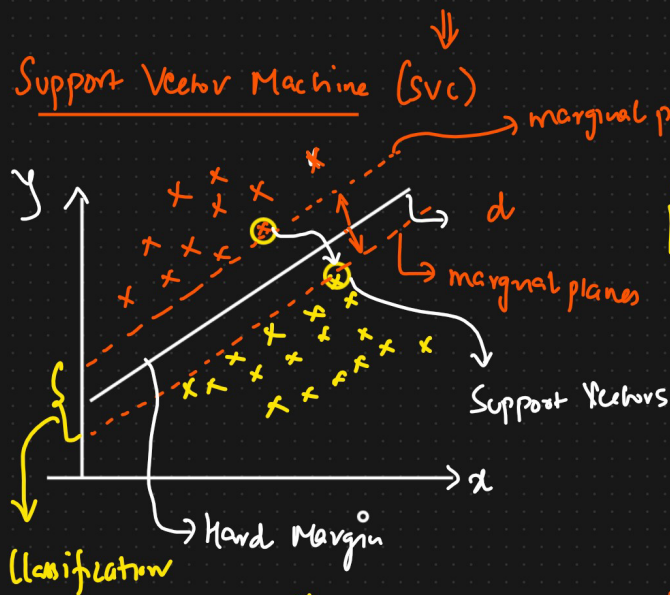
Support Vector Machines ML Algorithms.

① SVC (Support Vector Classifier)

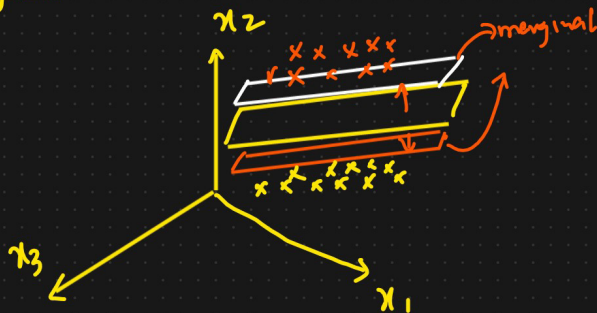
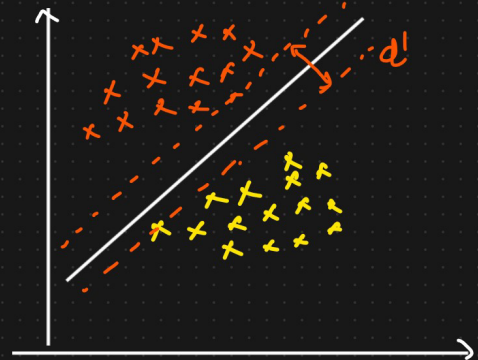
② SVR (Support Vector Regressor)



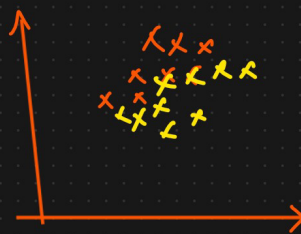
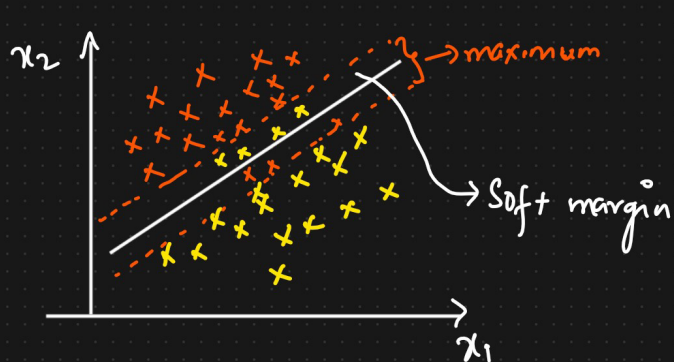
① Support Vector Machine (SVC)



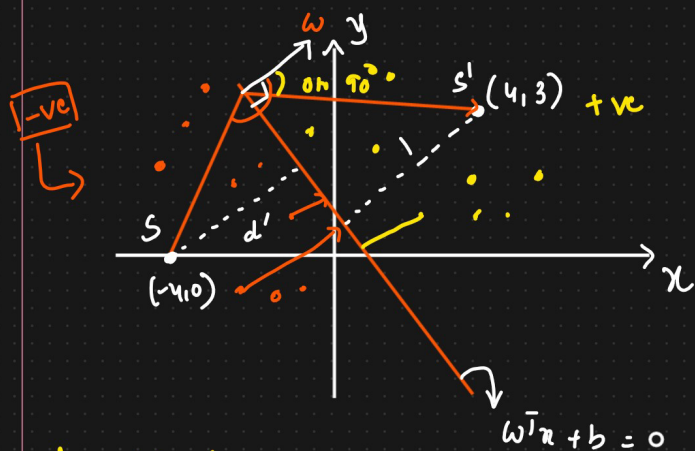
$$|d| > |d'|$$



Soft Margin And Hard Margin In SVM



Support Vector Machines (SVC) Maths Intuition



$$ax + by + c = 0$$

↓

$$w_1 x_1 + w_2 x_2 + b = 0$$

$$\boxed{w^T x + b = 0}$$

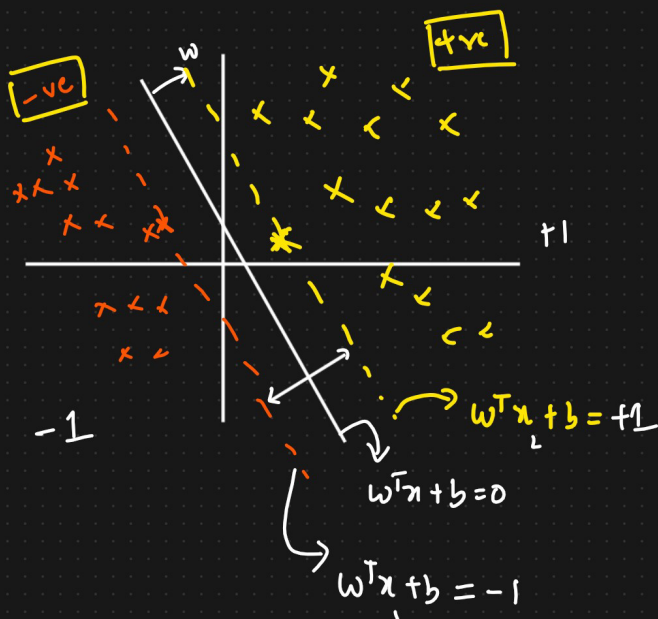
↓

$$b = 0$$

$$\boxed{w^T x = 0}$$

$d = -ve$ below plane

$d = +ve$ above plane



$$w^T x_1 + b = 1$$

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$$\begin{matrix} (-) & (-) & (+) \end{matrix}$$

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Unit vector { Magnitude of the vector is 1 }

Cost function

Maximize $\frac{2}{\|w\|}$ $\Rightarrow$ Distance between Marginal plane
 w, b

Constraint such that

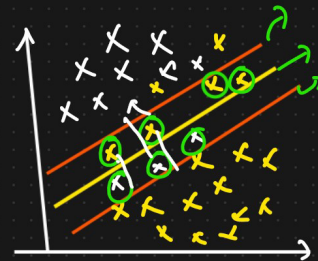
$$y_i \begin{cases} +1 & w^T x + b \geq 1 \\ -1 & w^T x + b \leq -1 \end{cases}$$

For all correct points

Constraint $\rightarrow y_i * (w^T x + b) \geq 1$

Maximize $\frac{2}{\|w\|}$ $\Rightarrow$ Min $\frac{\|w\|}{2}$
 w, b

$C_i = 6$ ✓



Cost function of SVM (svc)

Min $\frac{\|w\|}{2}$
 w, b

$+ \sum_{i=1}^n C_i \xi_i$

Hinge Loss

$\Downarrow$

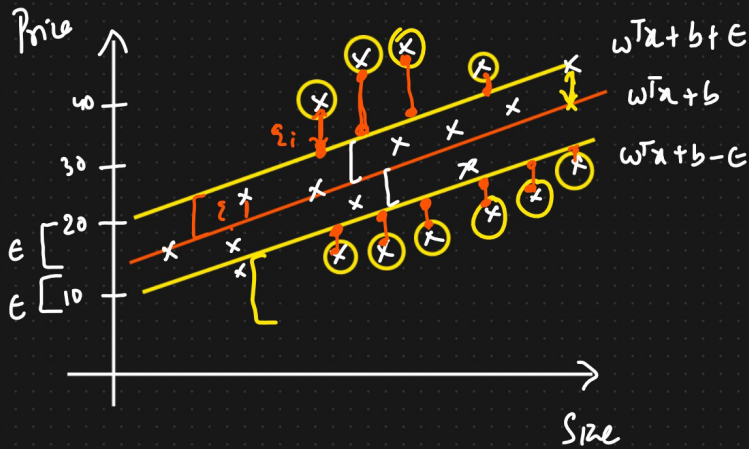
Soft margin

{ how many points we want to avoid misclassification }

{ distance of the incorrect data points from the marginal plane }

Support Vector Regression

ϵ : Marginal Error



Cost function

$$\text{Min}_{w, b} \frac{\|w\|}{2} + \underbrace{\left[C \sum_{i=1}^n \xi_i \right]}_{\text{hyperparameter}} \rightarrow \text{Hinge loss}$$

Constraint :

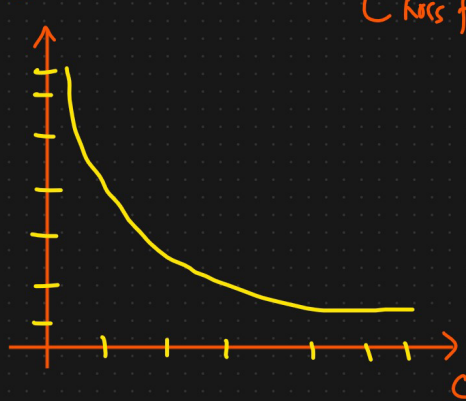
$$|y_i - w_i x_i| \leq \epsilon + \xi_i$$

$\Downarrow$
loss function

$\epsilon \rightarrow$ margin Error

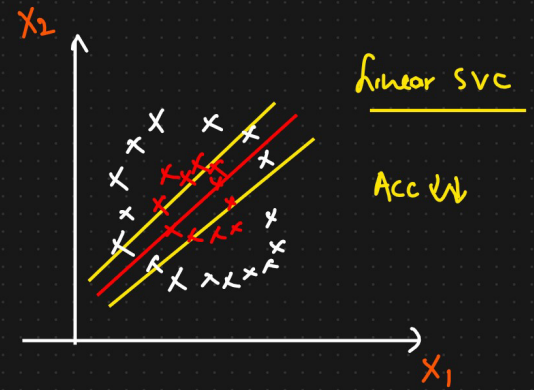
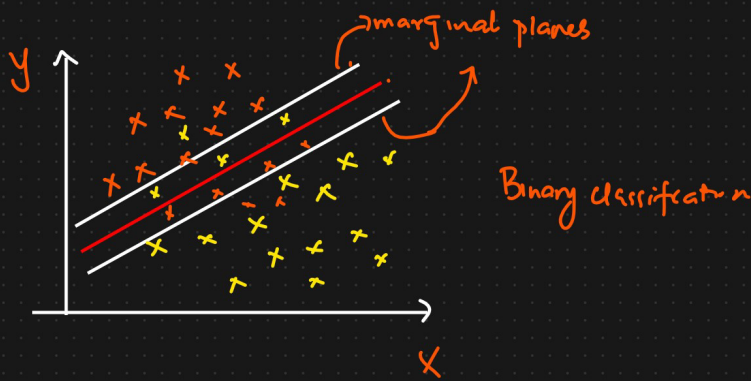
$\xi_i \rightarrow$ Error above the margin

loss function



Relationship
 $\left\{ \begin{array}{l} C \uparrow \\ \text{loss function} \downarrow \end{array} \right.$

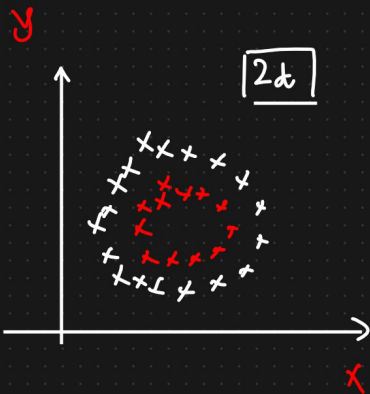
SVM KERNELS



Linear SVC



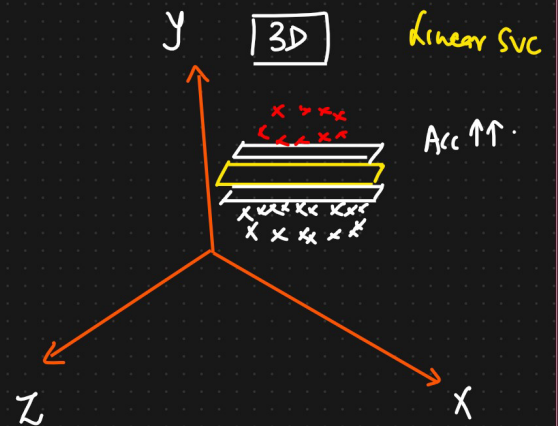
SVM Kernels



⇒ Transformations ⇒



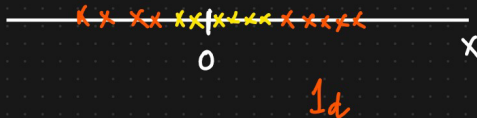
Mathematical
formulas



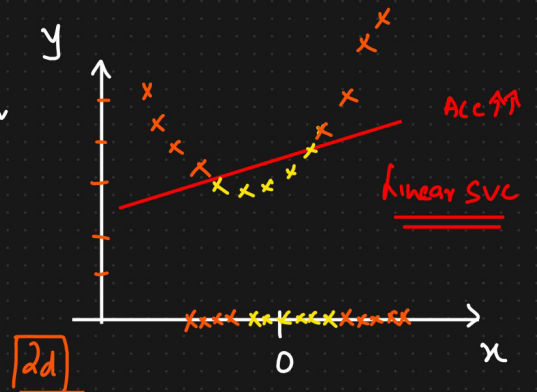
Dataset : 1d

SVM Kernel

Svc Linear SVC



⇒ Transformation
 $y = x^2$



① Polynomial Kernel

② RBF Kernel

③ Sigmoid Kernel